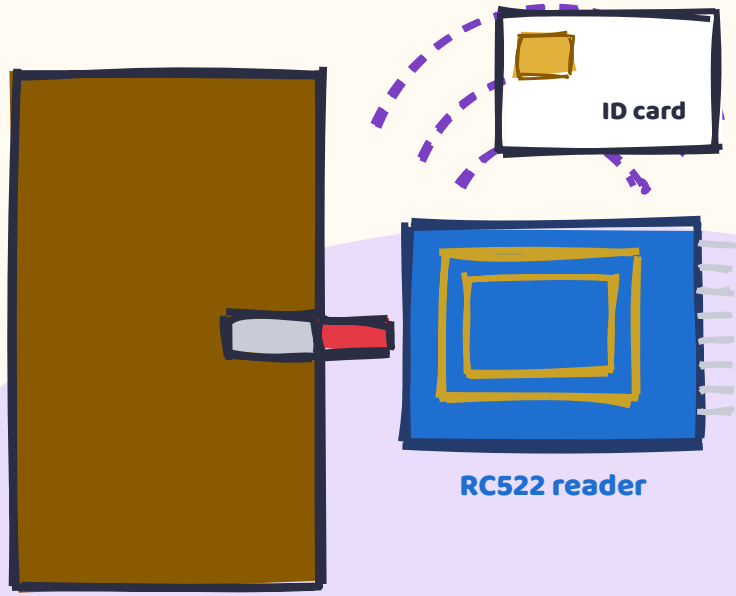
Young Innovators · Class 8

RFID Smart Door Lock with Access Log

Tap your card, the door opens. Strangers stay out!

RC522 RFID reader | Allow lists | Privacy & security



Did you know?
Most RFID cards have no battery. The reader's radio field powers the chip for a split second.

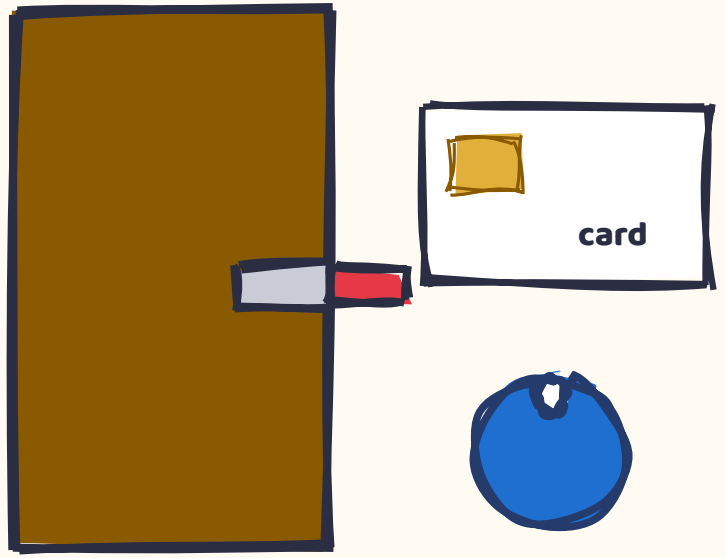
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Let's think...

What is wrong with metal keys?

Key problems

Keys get **lost** and copied. If a staff member leaves, you must change **every lock**. And a key can't tell you **who** opened the door, or **when**.



Hide the key?

Change all locks?

Use smart cards!

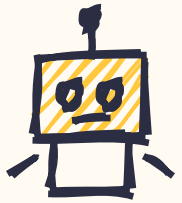
Today: you build a lock that only opens for **approved cards**, rejects strangers and keeps an **access log**.

Learning aims

Today I will learn to...

1. Explain how **RFID** cards work without batteries
2. Read a card's **UID** with the RC522 reader
3. Compare the UID with an **allow list** using arrays and loops
4. Unlock a **servo** and show an **access log** on the OLED
5. Discuss **privacy**, lost cards and how secure RFID really is

Keywords
RFID · UID · hexadecimal · array · allow list · log · privacy



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Watch and think

Video time!



What is RFID & How RFID Works

Louis Sirico · about 5 minutes · youtu.be/MAA9JpGraoU

Watch for...

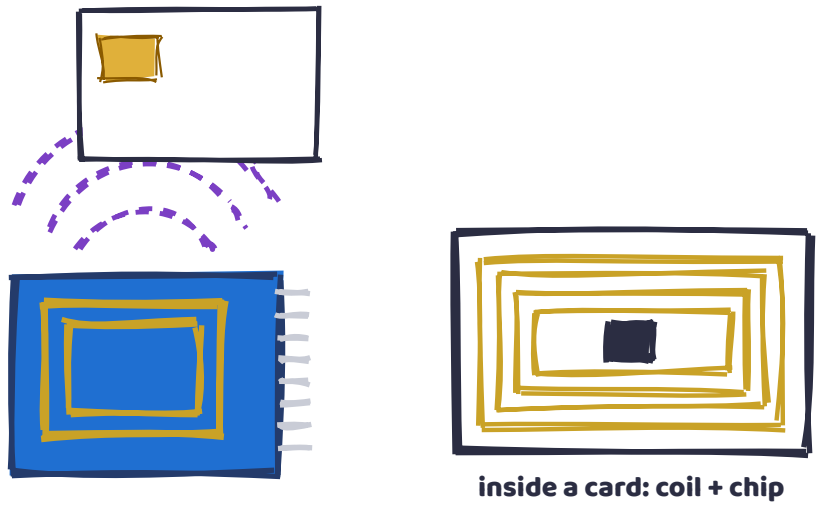
1. What are the two main parts of an **RFID** system?
2. How can a tag work with **no battery**?
3. Name two places RFID is used.

Scan to watch

After the video: Is RFID safe for a door lock? Discuss one strength and one weakness with your partner.

The science

How a card talks without a battery



inside a card: coil + chip

Step by step

1. The reader makes a **radio field** (13.56 MHz).
2. The card's **coil** catches energy, like a tiny wireless charger.
3. The chip wakes up and sends back its **ID number**.
4. All in a fraction of a second!

Short range on purpose

Our cards only work within a few centimetres. That stops the door opening when you just walk past.

RFID means

Radio Frequency IDentification. NFC on phones is a close cousin.

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Real world

Where do we tap cards?



Offices & hotels
Room key cards that can be switched off at reception



Metro & buses
Tap a travel card at the gate



Contactless pay
Bank cards and phones use NFC



Pets & farms
A tiny tag under the skin identifies lost pets

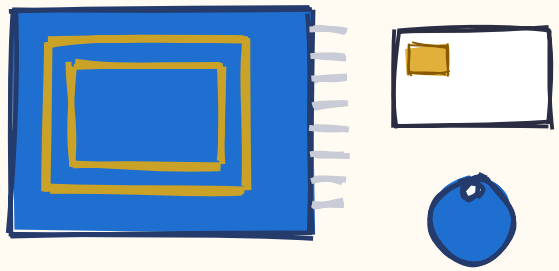
Which cards or tags do your family use? Do they say WHO you are, or just let you in?

Library books
Many libraries put RFID tags in books for quick self-checkout.

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Meet the reader

The RC522 RFID module



3.3V	3.3V (NOT 5V!)
RST	pin 9
GND	GND
MISO / MOSI	12 / 11
SCK	13
SDA (SS)	pin 10

3.3 V only!

Like the NRF24 radio, the RC522 is powered from **3.3V**. 5V can damage it.

SPI again

It uses the same **SPI** pins (11, 12, 13) as the radio in YI-10. **SDA** here means chip-select, not I2C!

Library

Install **MFRC522** (by GithubCommunity / miguelbalboa) from the Library Manager.

Cards + fobs

The kit comes with a white card and a blue key fob. Both work the same way.

Big idea

Every card has an ID called a UID

A3 4F 1B 9C

4 bytes, written in **hexadecimal**

Unique-ish

Cheap cards have 4-byte UIDs: about 4 billion possible numbers. Collisions are rare, but copies are possible (see slide 16!).

Hex digit	Normal number
0 to 9	0 to 9
A	10
B, C, D, E	11, 12, 13, 14
F	15
FF	255 (biggest byte)

Why hex?

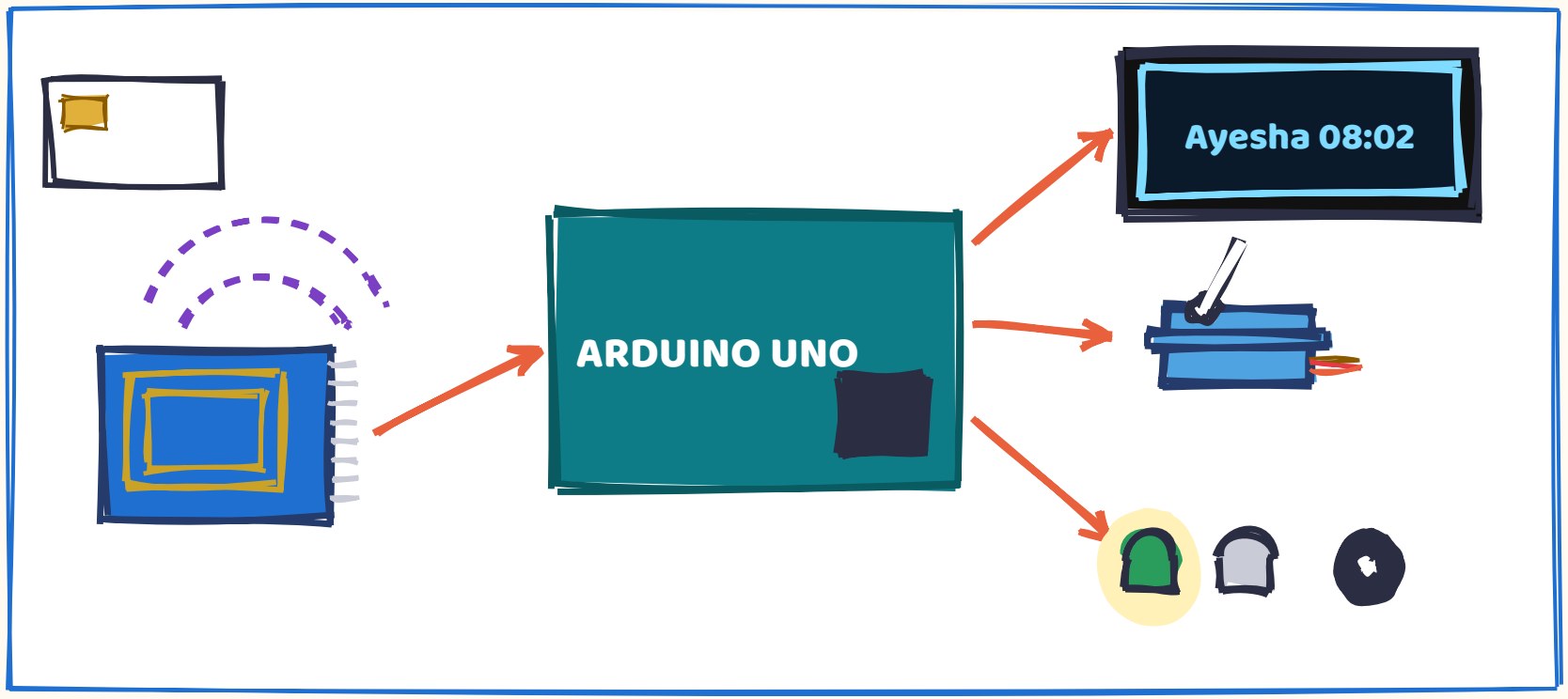
One byte (0 to 255) always fits in exactly **2 hex digits**. Easy to read and compare!

Try it

What is 2F in normal numbers? (2 × 16 + 15)

The plan

Read, check, decide, log



Two outcomes

Approved: green LED, servo unlocks for 3 s, name in log.

Unknown: red LED, 2 low beeps, UID logged as 'unknown'.

Servo = bolt

0° locked, 90° open. Fix a card 'bolt' to the horn.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. How safe is our lock really?
2. The bouncer
3. Our smart lock kit (per group)
4. Wiring the smart lock
5. Find your card's UID
6. The allow list and the bolt
7. An access log on the OLED

Think like a security engineer

How safe is our lock really?

UIDs can be copied: cheap 'magic' cards can clone a UID. Banks and offices use encrypted cards instead.

Lost card: remove its UID from the list straight away. No need to change the lock!

The log is personal data: it shows where people were and when. Who should be allowed to read it?

Fail safe: in a fire, should the door unlock automatically? Real buildings say yes.

Be fair: tell people that entries are logged, and delete old logs.



Discuss

Should a school log every time a student enters the library? Pros and cons?

Class game

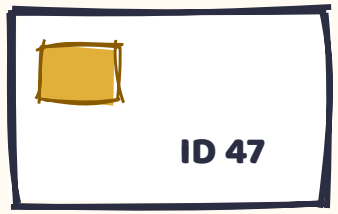
The bouncer

How to play

1. One student is the **bouncer** holding the secret **allow list** of 4 ID numbers.
2. Everyone else gets a card with a number.
3. Show your card. On the list? Come in, and the bouncer writes your name and time in the **log**.
4. Not on the list? Rejected!
5. Then: someone copies a valid number onto a blank card…

Then discuss

Could the bouncer tell the copied card from the real one? What does that mean for our lock?



Big idea
Checking an ID number is only as safe as the number is hard to copy.

Build it

Our smart lock kit (per group)

Arduino Uno
The brain

RC522 + card + fob
The reader

Servo SG90
The bolt

OLED 0.96 in
Access log

Green + red LED
With 2 × 220 Ω

Buzzer
Reject sound

Also: breadboard, M-F jumpers for the RC522, a card box door for the model, laptop with Arduino IDE 2.

Bring cards!
Old hotel keys or metro cards may also work, if they use 13.56 MHz.

MAKE

Build it

Wiring the smart lock

Part	Uno
RC522 3.3V / GND	3.3V / GND
RC522 SDA / RST	10 / 9
RC522 SCK / MOSI / MISO	13 / 11 / 12
Servo signal (orange)	pin 3
Servo red / brown	5V / GND
Green / red LED (+ 220 Ω)	pin 4 / pin 5
Buzzer +	pin 8
OLED SDA / SCL	A4 / A5

Check before USB

RC522 on **3.3V**? Servo on **5V**? Swapping them is the most common mistake.

Servo jitter?

If the Uno resets when the servo moves, power the servo from a separate 4×AA pack (share GND!).

MAKE

Code it: part 1

Find your card's UID

```
#include <SPI.h>
#include <MFRC522.h>
MFRC522 rfid(10, 9);            // SS (SDA), RST

void setup() {
  Serial.begin(9600);
  SPI.begin();
  rfid.PCD_Init();
  Serial.println("Tap a card...");
}

String readUID() {
  String id = "";
  for (byte i = 0; i < rfid.uid.size; i++) {
    if (rfid.uid.uidByte[i] < 0x10) id += "0";
    id += String(rfid.uid.uidByte[i], HEX);
    if (i < rfid.uid.size - 1) id += " ";
  }
  id.toUpperCase();
  return id;
}

void loop() {
  if (!rfid.PICC_IsNewCardPresent()) return;
  if (!rfid.PICC_ReadCardSerial()) return;
  Serial.println(readUID());
  rfid.PICC_HaltA();             // stop reading this card
}
```

Do this first!

Upload, open the **Serial Monitor**, tap each card and fob. **Write down** every UID. You need them for the allow list.

Why add a 0?

Byte 0x0A would print as just **A**. Adding the 0 makes it **0A**, so every UID has the same format.

Code it: part 2

The allow list and the bolt

```
#include <Servo.h>
Servo bolt;
String allowed[] = {"A3 4F 1B 9C", "04 7A 22 B1"};  // YOUR UIDs
String names[]   = {"Ayesha",      "Bilal"};
const int COUNT = 2;

// in setup(): bolt.attach(3); bolt.write(0);

// in loop(), after reading the UID:
String id = readUID();
int who = -1;                          // -1 = not found
for (int i = 0; i < COUNT; i++) {
  if (id == allowed[i]) who = i;
}
if (who >= 0) {
  digitalWrite(4, HIGH);               // green
  bolt.write(90);                      // unlock
  delay(3000);
  bolt.write(0);                       // lock again
  digitalWrite(4, LOW);
} else {
  digitalWrite(5, HIGH);               // red
  tone(8, 200, 300); delay(400); tone(8, 200, 300);
  delay(1000);
  digitalWrite(5, LOW);
}
```

Two arrays, same order

allowed[0] belongs to **names[0]**. The loop finds the position **i** of the matching UID, so we know the name too.

Why -1?

Positions start at 0, so -1 can never be a real position. It means 'not found'.

MAKE

Code it: part 3

An access log on the OLED

```
String logNames[3] = {"-", "-", "-"};
unsigned long logTime[3] = {0, 0, 0};

void addLog(String who) {
  for (int i = 2; i > 0; i--) {          // shift old entries down
    logNames[i] = logNames[i - 1];
    logTime[i] = logTime[i - 1];
  }
  logNames[0] = who;
  logTime[0] = millis() / 60000;         // minutes since power on
  showLog();                              // draw 3 lines on the OLED
}
```

```
ACCESS LOG
Ayesha 12 min
UNKNOWN 9 min
Bilal 3 min
```

Newest on top

Each new entry pushes the others **down**. The oldest falls off the end. This is called a **rolling log**.

No clock?

The Uno does not know the real time. We show **minutes since power on**. A real-time clock module would fix this.

Unknown cards too

Call addLog("UNKNOWN") for rejected cards. Security staff want to know!

Test like an engineer

Does the lock work?

Test	Expect	Result
Approved card	green, unlock 3 s, name logged	pass / fail
Approved fob	same as above	pass / fail
Unknown card	red, 2 beeps, UNKNOWN logged	pass / fail
Hold a card there for 5 s	only ONE unlock	pass / fail
Card 6 cm away	nothing	pass / fail

Debug checklist

No UID shown? Check 3.3V, SDA on 10 and RST on 9.
Approved card rejected? Copy the UID exactly, spaces and capitals included.

Print the UID AND the comparison result to the Serial Monitor while testing.

Level up!

Smart lock challenges

Choose your challenge

Master card: tapping the master card, then a new card, adds the new card to the list.

Lockout: 3 unknown cards in a minute = ignore all cards for 30 seconds and sound an alarm.

Remember: save the allow list in EEPROM so new cards survive a power cut.

Time rules: some cards only work during school hours (needs a real-time clock module).

Think like an engineer

Our list lives inside the lock. How do big buildings with 500 doors update every lock when someone leaves?

Boss level

Locks connect to a central server over the network. One change updates every door!

Show what you know

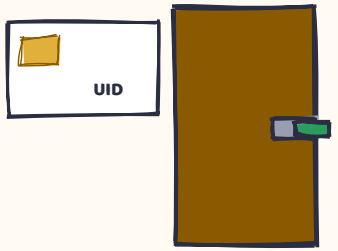
What did we learn?

- [x] RFID cards get power from the reader's radio field
- [x] Each card has a UID, shown in hexadecimal
- [x] An allow list + loop decides who may enter
- [x] A rolling log records who entered and when
- [x] UID-only locks can be cloned, and logs are personal data

Exit ticket: why do we use -1 to mean 'not found'?

At home

Count how many RFID or NFC cards your family uses. What does each one open or pay for?



Next class: a Rechargeable Power Pack for Robots!